

Api 510 Pressure Vessel Inspection

Page 1: Scope, Inspection Frequencies, and MAWP

API 510 is the international Pressure Vessel Inspection Code governing in-service inspection, repair, alteration, and re-rating of pressure vessels. It applies to vessels in the petroleum, petrochemical, and chemical processing industries that have been placed in service.

External inspections must be performed every 5 years or at the same interval as the internal/on-stream inspection, whichever is less. Internal or on-stream inspections must be performed at the half-life of the vessel ($\text{Remaining Life} / 2$) or every 10 years, whichever is shorter. The Maximum Allowable Working Pressure (MAWP) must be recalculated based on measured wall thickness.

Page 2: Corrosion Rate and Remaining Life Calculations

The corrosion rate (CR) must be calculated to estimate remaining life. The short-term corrosion rate is: $CR = (t_{\text{previous}} - t_{\text{actual}}) / \text{time_interval}$. The long-term corrosion rate is: $CR = (t_{\text{initial}} - t_{\text{actual}}) / \text{time_interval_in_service}$.

The remaining life (RL) of the vessel in years is calculated as: $RL = (t_{\text{actual}} - t_{\text{min}}) / CR$, where t_{actual} is the measured thickness at the thinnest point, and t_{min} is the minimum thickness required for the design pressure and temperature. Thickness measurements must be recorded at designated Thickness Monitoring Locations (TMLs).